MIL-M-36727B
23 September 1978
SUPERSEDING
MIL-M-0036727A(DLA-DM)
12 May 1977, and
MIL-M-36727
10 December 1970

MILITARY SPECIFICATION

MIRROR, AND MIRROR-LIGHT SET, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mirror, and a mirror-light set, suitable for use by a dental patient in monitoring his oral hygiene practices by self-examination of the dental and oral structures.

1.2 Classification. The mirror and mirror-light set covered by this specification shall be of the following types, as specified (see 6.1):

Type I. - Mirror, Plastic.
Type II. - Mirror-Light Set.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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SPECIFICATIONS

FEDERAL

L-P-390	- Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, And High Density).
L-P-1183	- Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS), Rigid.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-46129	- Plastic, Molding And Extrusion Material, Polyphenylene Oxide, Modified.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24179	- Mirror-Light, Dental, 115 Volt, 60 Cycle, AC.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 544 - Standard for Electric Medical and Dental Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062, 1285 Walt Whitman Road, Melville, LI, NY 11746, 1655 Scott Blvd., Santa Clara, CA 95050.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Plugs and Receptacles, General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standard of Safety No. 544. Proof shall consist of one of the following:

A. UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, or the Government Inspector, upon request.

B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test

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methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as, materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts lists, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specification.

3.2 Materials. Materials shall be new and shall conform to the applicable specifications and requirements specified herein. Materials not specifically designated shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended.

3.2.1 Type I mirror. The type I mirror shall be fabricated of high impact, Acrylonitrile-Butadiene-Styrene (ABS) conforming to type 2 of L-P-1183 (see 6.3). The mirror shall have a composite electroplating of chromium over nickel.

3.2.2 Type II mirror-light set. The type II mirror-light components shall be fabricated of the materials shown in table I.

TABLE I. Materials.

Components	Materials
Swivel arm, transformer and light source housings	Modified polyphenylene oxide thermoplastic resin, class 2 of MIL-P-46129 (see 6.2)
Switch trigger	High impact phenolic nylon
Lamp socket	Brass
Rivets, washers	Nickel-plated brass
Screws	Plated steel
Insulation board	Vulcanized fiber
Light source reflector and outer surface	ABS with chromium over nickel electroplating, type 2 of L-P-1183 (see 6.3)
Mirror	Polished glass coated with a silver-copper plating and mirror paint backing
Mirror frame	Polyethylene, type I, class L, grade 2 of L-P-390

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3.3 Style, design, and dimensions. The type I mirror and the type II mirror-light set shall conform to Defense Personnel Support Center (DPSC) Drawing No. 24179 for style, design, and dimensions.

3.3.1 Type I mirror. The type I mirror shall be one piece and all surfaces of the mirror shall have a 14-inch spherical radius. The type I mirror shall be an intra-oral mirror and shall permit a dental patient to self-examine and obtain a clear, undistorted image of the otherwise hidden areas of his dental and oral structures when tested as specified in 4.5.2.

3.3.2 Type II mirror-light set. The type II mirror-light shall consist of a set including a high intensity light, the 5-inch magnifying mirror, and three type I intra-oral mirrors. The mirror-light shall be suitable for use by a dental patient in the self-examination of his dental and oral structures. The transformer shall step down the 120 volt circuit voltage to 6 secondary volts and shall be enclosed in a gray plastic housing. The light source housing shall be plated on all surfaces and shall contain a bayonet-type socket equipped with a No. 55 bulb (6240-00-012-5588). The light source housing, swivel arm ends, and transformer housing shall be opaque light gray or beige in color, and shall be equipped with clevises which shall permit the three members to be pinned together with rivets and provide 170 $\pm$ 10 degree ($^{\circ}$) rotation of both the swivel arm and the light source. The riveted connections shall be sufficiently tight to insure friction-locking of the light source in any desired operating position. A slot shall be provided in the front surface of the transformer housing to permit the entire lamp unit to be hung from a picture hook. An "on-off" switch shall be provided in the front surface of the transformer housing. The total weight of the 5-inch mirror, light source, swivel arm, and transformer housing shall be 17 ounces. The mirror-light shall be provided with an eight-foot long, 3 wire electric cord with a 3 prong plug on one end suitable for use with receptacles specified in ANSI C73.11. The type II light unit shall withstand, without failure, the dielectric strength test specified in 4.5.3.

3.3.2.1 Glass mirror. The mirror shall consist of a 5-inch magnifying glass mirror, plated as specified in table I and mounted in a detachable, gray, plastic frame. The frame shall be provided with a resilient sleeve adaptor suitable for attachment to the light source housing of the standard Army flashlight, NSN 6230-00-264-8261. The sleeve adaptor shall be provided with an individual stop at bottom and a back-of-frame stop at top to limit the seating of the frame on the flashlight, and shall have a rib on a 30° radial line in each lower quadrant. The glass mirror frame, when properly positioned on the flashlight, shall be at right angles to the direction of the light beam. The glass mirror shall have an 18-inch spherical

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radius which shall permit the focusing of the light beam on the patient's mouth when held at distances up to 7 inches from the lips. This mirror shall comply with the assembly and reflection test requirements as specified in 4.5.1 and 4.5.2, respectively. Total weight of the 5-inch frame and mirror shall be 3 ounces.

3.4 Instructions. A set of instructions shall be included with each unit of issue for type I mirror and type II mirror-light set. Instructions shall include pertinent information concerning use of mirrors, parts list and cost data, and method of adjustment of mirror to flashlight (6230-00-264-8261) and mirror-light (type II).

3.5 Tolerances. Unless otherwise specified, tolerances on weights shall be ± 5 percent. Function and highest quality workmanship shall not be sacrificed.

3.6 Finish. The type I mirror shall have a polished mirror finish on its concave surface and a dull or matte finish on the reverse side (convex surface) to distinguish it from the reflector surface. This type I mirror shall have an otherwise smooth surface finish to prevent any irritation to the mouth tissues. The light source reflector surface of the type II mirror-light unit shall have a highly polished, mirror finish. The mirror of the type II mirror-light set shall be of such quality and finish as to provide a highly reflective dental image during self-examining procedures. All frames and housings shall be free from sharp or rough edges. All plating shall be adherent and free from blisters or other defects that may cause peeling.

3.7 Identification marking. The transformer housing rear surface and cover, and the back of each mirror frame shall be permanently and legibly marked with the name or registered trademark of the manufacturer. In addition, the transformer housing cover shall be permanently and legibly marked with the transformer input-output data and with light bulb (6240-00-012-5588) manufacturer's number.

3.8 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall elect dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit, and function. No deviations from dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

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3.9 Workmanship. Workmanship shall be first class throughout. The mirror and mirror-light set shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the materials used in the mirror may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been shall maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for examination purposes shall be one mirror or one mirror-light set, as applicable.

TABLE II. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Minor A	II	2.5
Minor B	S-2	4.0

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. The unit of product for test purposes shall be one mirror or one mirror-light set, as applicable.

TABLE III. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Mirror (type I)	Reflection test	3.3.1	4.5.2	II	1.5
Mirror of type II mirror-light set	Reflection test	3.3.2.1	4.5.2	II	1.5
Mirror of type II mirror-light set	Assembly test	3.3.2.1	4.5.1	S-2	1.5
Mirror-light set (type II)	Dielectric test	3.3.2	4.5.3	S-2	1.0

4.4 Examination. The mirror and mirror-light set shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table IV. Examination shall not be restricted to the classified possible defects listed below.

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TABLE IV. Classification of defects.

Categories	Defects
Major A:	
101	Plating peeling or flaking.
102	Reflective surfaces not free from waviness.
103	Style or design not as specified.
104	Components of mirror-light missing.
105	Glass mirrors not adequately secured in frames.
106	Finish of reflective surfaces not as specified.
107	Assembled parts not free from cracks, burrs, or rough or sharp edges.
Minor A:	
201	Finish of non-reflective surfaces not uniform or not as specified.
202	Instructions for use not as specified.
203	Identification marking not present or not legible.
Minor B:	
251	Dimensions not as specified.
252	Weight not as specified.

4.4.2 Packaging inspection. The inspection of the unit package, intermediate package, packing, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Assembly test. Plastic or metallic pieces of tubing, probes, or inserts, having individual diameters of 2.02 and 2.05 inches, shall be alternately pressed into the adaptor sleeves of the frames of the glass mirror of the type II mirror-light set to check that these parts will suitably fit the light source and flashlight housings referenced by this specification. This test shall be supplemented by an assembly test performed by laying the mirror frame face downward on a flat surface with the vertical axis of the mirror aligned in the position to be assumed on the light housing. The light housing shall then be pressed uniformly into the

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sleeve adaptor to the depth permitted by the adaptor stops. The position of the mirror frames on the light housings shall then be checked for (1) seating of housing against frame stops, (2) ease of repositioning, and (3) vertical alignment of mirror. Any adaptor sleeve which will not support the mirror frame in the positions defined in 3.3.2.1 shall be rejected.

4.5.2 Reflection test. A reflection test shall be performed on the type I mirror and the glass mirror of the type II mirror-light set by holding a sheet of printed matter, facing away from the operator, in the left hand or in a stand 10 ± 2 inches from the face. The right hand shall move the mirror parallel and 1 ± 1/2 inches behind the sheet of printed matter, mirror side to the operator. The print shall be observed in the mirror.

4.5.3 Dielectric test. A dielectric test of 1250 volts, 60 cycle, single phase, alternating current shall be applied to all type II light units, for one minute, between the exposed non-current carrying parts and the current carrying parts, with all current parts in the circuit.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing ten type I mirrors or one type II mirror-light set, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging.

5.1.2.1.1. Type I. Each mirror shall be inserted into a kraft envelope of appropriate size and the envelope sealed. Ten envelopes shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2.1.2 Type II. Each mirror-light set shall be wrapped in neutral tissue paper and packaged in a box conforming to PPP-B-636, class domestic. The box shall include suitable interior fitting or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure of the box shall be as specified in the appendix of the box specification.

5.1.2.2 Intermediate package. Twelve unit packages containing type I mirrors or type II mirror-light sets shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

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5.1.3 Level C. Each type I mirror or each type II mirror-light set shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (144 units) of type I mirrors or two intermediate packages (24 units) of type II mirror-light sets shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages (144 units) of type I mirrors or two intermediate packages (24 units) of type II mirror-light sets shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged mirrors or mirror-light sets shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information on the procurement documents:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Modified polyphenylene oxide thermoplastic resins suitable for use in the mirror-light are General Electric Co.'s "Noryl", Grade SE1-802, or Marbon Chemical Co.'s Cycolac KJB.

6.3 An ABS resin suitable for use for the mirror-light housing reflector and for the type I mirror is Marbon Chemical Co.'s Grade EP-3510.

6.4 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation
6520-00-019-9268	MIRROR, MOUTH EXAMINING, Patient-training, Plastic, 10s	Type I
6520-00-027-5869	MIRROR-LIGHT, DENTAL, 115 volt, 60 Hz, AC	Type II (includes 5-inch glass mirror)

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6520-1470

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-36727B MIRROR, AND MIRROR-LIGHT SET, DENTAL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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